6.9 Integrands Containing $ax^2 + bx + c$; Integrate Using Long Division

Name _____ Period ____ Date _____

AP Calculus 6.9 · Student sheet

I can…

- complete the square in a quadratic.
- integrate 1/(quadratic) with an inverse tangent.
- use long division before integrating an improper rational function.

Vocabulary · fill in the missing word

Completing the square — _____ $x^2 + bx + c$ as $(x + \frac{b}{2})^2 + (c - \frac{b^2}{4})$

Improper rational function — a rational function whose numerator has degree ≥ the degree of its _____

Part 1 · Complete the square

Write each as $(x + h)^2 + k^2$.

Quadratic	$(x + h)^2 + k^2$
$x^2 + 4x + 5$	
$x^2 - 6x + 13$	
$x^2 + 2x + 10$	

Part 2 · Notes and examples (fill in each note, then work the example)

Complete the square: $x^2 + bx + c = (x + b/2)^2 + (c - $ _____$)$.

Inverse tangent: $\int du/(u^2 + k^2) = (1/k)$_____$(u/k) + C$.

Example 1

Find $\int dx/(x^2 + 4x + 5)$.

Example 2

Find $\int dx/(x^2 - 6x + 13)$.

Improper rational function: use long _____ first.

Example 3

Find $\int \frac{x^2 + 1}{x + 1}\,dx$.

Example 4

Evaluate $\int$ from -2 to -1 of $dx/(x^2 + 4x + 5)$.

Part 3 · Partner practice

Partner 1

Find $\int dx/(x^2 + 2x + 2)$.

Partner 2

(AP® Practice) $\int dx/(x^2 + 6x + 13) =$

(A) $(\frac{1}{2})\tan^{-1}(\frac{x + 3}{2}) + C$ (B) $\tan^{-1}(x + 3) + C$

(C) $\ln(x^2 + 6x + 13) + C$ (D) $2\tan^{-1}(\frac{x + 3}{2}) + C$